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Nematicidal activity of *Streptomyces* sp. GHRS-5 against root-knot nematode *Meloidogyne enterolobii* in guava

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Abstract

The root-knot nematode, *Meloidogyne enterolobii* has emerged as a major global threat, particularly impacting guava production in India. Effective management of this invasive species requires sustainable alternatives to synthetic nematicides due to concerns over environmental safety and nematode resistance. *Streptomyces* spp. metabolites offer a promising solution by producing bioactive compounds that suppress nematodes while enhancing plant defenses. *In vitro* bioassays identified GHRS-5 as the most potent strain, showing significant inhibition of egg hatching and increased juvenile mortality rates over 24,48 and 72 hours. Metabolic profiling using GC-MS revealed nematicidal biomolecules which were further analyzed through molecular docking studies. These studies demonstrated strong binding affinities between effective biomolecules and key nematode proteins including cellulose binding protein, cytochrome oxidase subunit I, chorismate mutase, pectate lyase, and major sperm protein suggesting potential mechanisms for nematode suppression. Pot experiments confirmed the efficacy of GHRS-5 with an 74.5% increase in root length, a 60% decline in female nematodes, and a 65% decrease in soil populations. These results highlight that GHRS-5 plays a dual role in boosting plant growth as well as nematode suppression. This eco-friendly approach offers a sustainable solution for guava farmers and will pave the way for broader application in managing *M. enterolobii* across crops. Future research will focus on scaling these findings to field conditions and explore broader applications.

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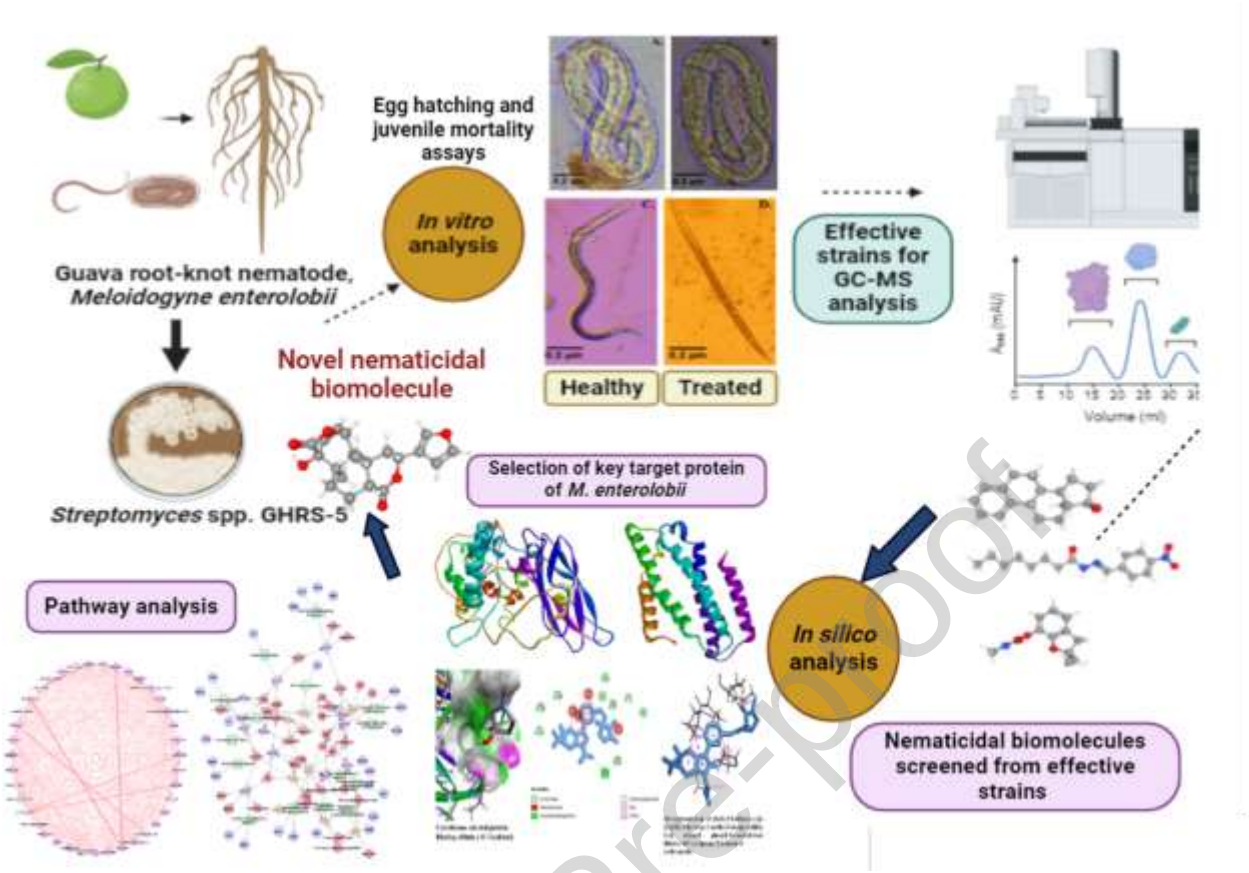
Graphical Abstract



Keywords:

Guava cultivation; *Streptomyces* metabolites.; GC- MS analysis; Molecular modelling and docking; pot experiments

1. Introduction

Nematodes, commonly known as roundworms are one of the most diverse groups of animals on earth (Blaxter 2009). They occupy a wide range of ecological niches, from free-living organisms in soil and water to parasites of plants. Nematodes play a crucial role in nutrient cycling and soil health, but some species can cause significant damage to agricultural systems. Plant-parasitic nematodes (PPNs) are a major threat to global agriculture, with the root-knot nematodes (*Meloidogyne* spp.) being some of the most destructive. These nematodes infect plant roots, forming galls and knots that disrupt water and nutrient uptake, reducing crop yields. Among the various plant-parasitic nematodes, *Meloidogyne enterolobii* has emerged as a particularly aggressive species due to its wide host range, ability to overcome crop resistance

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genes, and rapid reproduction rate. The guava root-knot nematode, *Meloidogyne enterolobii* has become a significant threat to guava farmers in India leading to a 65% reduction in annual yields equivalent to a financial loss of 242.1 billion Indian rupees under open field conditions (Sinha 2021). Its global distribution, broad host range, and ability to reproduce on crops with resistant genes underscore the need for a focused search on this invasive species. *M. enterolobii* is recognized as a major pest in Brazil, South America, Asia, and sub-Saharan Africa. Also, several countries have designated it a quarantine pest to curb its spread.

Management of nematodes requires a safer method to achieve sustainable agriculture than synthetic nematicides. Effective chemical nematicides often face challenges that limit their prominence and efficiency in nematode management (Fabiyi and Bello 2023). Over time, nematodes can develop resistance to certain chemical nematicides, making them less effective. The inefficiency and diminishing prominence of chemical nematicides are largely due to their environmental, health, and economic drawbacks. These factors and the emergence of resistance, have led to a shift toward safer more sustainable alternatives like biological control agents, including microbial-based products such as *Streptomyces* spp. and other integrated pest management strategies.

For instance, nematicides like methyl bromide have been phased out due to their harmful environmental impact, and alternatives like oxamyl and fluensulfone have limited efficacy against *M. enterolobii*. Additionally, the high cost of nematicides and their negative impact on beneficial soil organisms make them less desirable for long-term control. The combination of *M. enterolobii'*s wide host range, ability to overcome plant resistance, high reproductive capacity, and the limitations of current chemical and host-plant resistance methods make managing this nematode exceptionally difficult. As a result, the integrated pest management (IPM) approach is preferred, emphasizing bionematicides as a crucial component in nematode control. Studies have shown the potential of beneficial microorganisms in suppressing plant pathogens, including nematodes. These factors underscore the need for biological control methods.

At this instance, biological control of nematodes is gaining attention due to its reputation as an environmentally friendly approach for managing plant-parasitic nematodes. Also, controlling nematodes using bacterial metabolites has garnered increasing attention as an environmentally friendly and effective approach for PPN. These metabolites are often derived from bacteria such as *Streptomyces* spp. exhibit nematicidal properties offering a sustainable alternative to chemical nematicide. Bacterial metabolites are natural compounds produced by various bacterial spp. that significantly combat plant pathogens. These metabolites include antibiotics, enzymes, and lipopeptides which inhibit the growth of harmful fungi, bacteria, and nematodes. For instance, *Bacillus subtilis* produces lipopeptides like surfactin and iturin

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which disrupt fungal cell membranes while *Pseudomonas fluorescens* generates phenazines and hydrogen cyanide that inhibit pathogen growth (Raaijmakers *et al.,* 2010).

Bacterial metabolites have been extensively studied for their potential in controlling PPN offering an eco-friendly alternative to chemical treatments. Several bacterial genera, including *Bacillus, Pseudomonas*, and *Streptomyces* produce metabolites with nematicidal and nemastatic properties. Recent research has shown that various *Streptomyces* spp. isolates proven evident nematicidal effect against phytonematodes (Chen *et al.,* 2024). *Streptomyces* spp. have emerged as potent biocontrol agents against PPN due to their ability to produce a wide array of bioactive metabolites with nematicidal properties (Mani *et al.,* 2024). These soil-dwelling, filamentous bacteria are well-known for their diverse secondary metabolites, including antibiotics, enzymes, and volatile organic compounds which directly target nematodes and indirectly promote plant health by stimulating plant defenses. Apart from directly killing nematodes, *Streptomyces* metabolites can induce systemic resistance in plants. ISR primes the plant's immune system to respond more rapidly and robustly to nematode invasion. This results in enhanced plant defenses against nematodes and other pathogens, improving overall plant health. Identifying novel nematicidal biomolecules from biocontrol agents will facilitate effective nematode management. To accomplish this, virtual screening and molecular docking techniques can aid in discovering promising new bioactive compounds targeting specific nematode species and utilizing *Streptomyces* spp. biomolecules with antagonistic properties against PPN will contribute to successful nematode control (Chen *et al.,* 2024).

Building on this, our current research explores the potential of *Streptomyces* spp. as a natural nematicidal agent specifically targeting *M. enterolobii.* This research aims to identify novel biomolecules produced by *Streptomyces* spp. that exhibit potent antagonistic effects for control. Our investigation involves screening different *Streptomyces* isolates for their ability to suppress nematode populations studying their mechanisms of action and evaluating their effectiveness under controlled and field conditions. By leveraging the diverse secondary metabolites produced by *Streptomyces* such as enzymes and volatile organic compounds, we aimed to develop an eco-friendly solution for nematode management. This research holds promise for revolutionizing the management of *M. enterolobii* infestations offering a sustainable alternative to chemical nematicides and contributing to improve agricultural productivity.

2. Materials and methods

2.1. Nematode culture

Guava roots infested with the root-knot nematode, *M. enterolobii* were collected from various guava growing regions in Coimbatore and Erode districts of Tamil Nadu, India. The roots were thoroughly washed to remove soil particles, and egg masses were aseptically collected under a stereo zoom microscope before being incubated at room temperature (28 ± 2°C, 80% RH) for three days (Mongae 2013). Freshly

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hatched 5000±80 second-stage juveniles (J_2) were inoculated into guava plants grown in the glasshouse at the Department of Nematology, TNAU. These roots served as the inoculum for further studies.

2.2. Morphological and molecular identification of *M. enterolobii*

Since the perineal patterns of *M. enterolobii* are quite similar to the morphological characteristics of *M. incognita*, molecular characterization was carried out to confirm the species (Decraemer *et al.,* 2003). A sterilized gravid female root-knot nematode was crushed and placed into a microfuge tube containing worm lysis buffer (0.2 M NaCl, 0.2 M Tris base, 1.5 µL beta-mercaptoethanol, and 1 µL Proteinase K). The tube was centrifuged and then incubated in a dry bath at 60°C for 3 hours, followed by heating at 99°C for 10 minutes in a thermal gradient cycler. The DNA was then stored at -20°C until further use. The DNA of *M. enterolobii* was used for PCR amplification with primers TW 81F (GTTTCCGTAGGTGAACCTGC) and AB 28R (ATATGCTTAAGTTCAGCGGGT) (Hu *et al.,* 2011). The PCR process included an initial denaturation at 94°C for 5 minutes, followed by 35 cycles of initial denaturation at 94°C for 5 minutes, followed by 35 cycles of denaturation at 94°C for 30 seconds, annealing at 54°C for 30 seconds, and extension at 72°C for 1 minute, with a final extension at 72 °C for 10 minutes. The PCR products (10 µL) were run on a 1% agarose gel in TAE buffer at 70V for 50 minutes and visualized under UV light using a Vilber Ebox system. The amplified DNA was then sent to Biokart Pvt Ltd., Bangalore for sequencing, and gene homology was analyzed using NCBI BLAST with the non-redundant nucleotide database (nr database).

2.3. Isolation and identification of *Streptomyces* spp. from healthy guava communities

Soil samples were collected from the rhizosphere regions of healthy guava crop plants across various localities in Coimbatore and Erode districts. These samples were serially diluted up to a 10^{-4} dilution, and 1 ml of each dilution was plated on Ken Knight's medium (KM) (Rahman *et al.,* 2010). The plates were incubated at 28 ± 2°C for 72 hours. *Streptomyces* spp. colonies were selected based on their morphological characteristics: aerobic, gram-positive, multicellular, and filamentous bacteria producing well-developed vegetative hyphae with branches. These colonies were further purified through repeated streaking on KM and incubated at 28 ± 2°C for subsequent studies.

The identity of antagonistic *Streptomyces* spp. was confirmed and amplified using universal primers 27F and 1492R. The PCR process involved an initial denaturation step at 94 °C for 3 minutes, followed by 40 cycles of denaturation at 95°C for 1 minute, annealing at 60°C for 2 minutes, and extension at 72°C for 10 minutes. The amplified PCR products were analyzed using 1% agarose gel electrophoresis and visualized with a gel documentation system to confirm successful amplification (Al_husnan and Alkahtani 2016). Subsequently, the PCR products were sequenced and the resulting sequences were compared for

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similarity using the BLASTN tool. The validated sequences were submitted to the NCBI GenBank database, and unique accession numbers were obtained.

2.4. Testing the nematicidal activity of isolated *Streptomyces* sp. strains

Isolated *Streptomyces* spp. strains were evaluated for their effectiveness in reducing the population of *M. enterolobii.* A single colony was inoculated into Ken Knight's agar broth and incubated on a shaker set at 180 rpm and 35°C for 5 days. The pure culture extract was collected by centrifuging the inoculated broth at 10,000 rpm for 15 minutes at 4°C. The supernatant obtained was then filtered using a 0.2 µm bacterial syringe filter. This filtered suspension was subsequently utilized to assess its effects on egg hatching and juvenile mortality (Nagachandrabose *et al.,* 2024).

2.5. Testing the nematicidal efficacy of *Streptomyces* sp. strain GHRS-5 against eggs of *M. enterolobii*

The antagonistic potential of isolated *Streptomyces* spp. was assessed through egg-hatching assays. The bioassay included 11 treatments consisting of 5 *Streptomyces* spp. isolates with concentration of 25%, 50%, 75%, 100% pure raw cell free culture filtrate per treatment, and two controls were kept. The controls contained broth without *Streptomyces* culture and plates with distilled water were maintained. Each treatment had five replications and two repetitions, with one plate representing a single replicate. Nematode egg suspensions were collected and tested for efficacy using 2.5 ml of pure culture extract 30mm×15mm plates containing 100 *M. enterolobii* eggs each were incubated at 28 ± 2°C (Nagachandrabose *et al.,* 2023). Egg masses were isolated from infected roots, placed in Petri plates with sterile distilled water, and incubated at 28 ± 2°C for 72 hours. Observations were recorded on the total number of hatched eggs on 24, 48, and 72 h.

2.6.Testing the nematicidal efficacy of *Streptomyces* sp. strain GHRS-5 against juveniles of *M. enterolobii*

For juvenile mortality testing, freshly hatched juveniles at a concentration of 100 juveniles per ml were placed in Petri plates containing the *Streptomyces* culture suspension and incubated at 28 ± 2°C. plates with broth alone and distilled water were served as control. Each treatment had five replications and two repetitions, with one plate representing a single replicate with a concentration of 25%, 50%, 75%, 100% per treatment. The number of dead juveniles was recorded at 24, 48, and 72 hours after inoculation. Experiments were conducted twice to confirm the results. Juveniles that remained straight and showed no movement when prodded were counted as dead. The *Streptomyces* spp. strains showing the most effective activity against the eggs and juveniles of *M. enterolobii* were selected for further testing.

2.7. Metabolic profiling of biomolecules from effective S*treptomyces* sp. strains

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Based on the *in vitro* screening results, *Streptomyces* spp. supernatants showing the highest antagonistic effects on egg-hatching inhibition and juvenile mortality were selected for secondary metabolite profiling. The biomolecules from *Streptomyces* sp. isolates were analyzed using Gas-Chromotography/ Mass Spectrometry (GC-MS). *Streptomyces* cultures were cultivated in Ken Knights' medium (KM) broth and incubated on a shaker at 180 rpm for 5 days. The culture broth was then centrifuged at 6000 rpm for 15 minutes, and the supernatant was carefully separated. The pH of the supernatant was adjusted to 2, and equal volumes of HPLC-grade ethyl acetate were added, followed by overnight shaking at 150rpm (Nandhini *et al.,* 2018). The mixture was transferred to a separating funnel, and the organic layer was collected and dried in petri dishes. After the solvent evaporated, the remaining residue was dissolved in 1 ml of HPLC-grade methanol. A one ml sample was then subjected to GC-MS analysis using the Perkin Elmer Clarus SQ8C system.

2.8. Molecular modelling of potential protein targets of *M. enterolobii*

Based on the literature survey, the protein targets, which include Cellulose binding protein, Major sperm protein, Cytochrome oxidase subunit I, Pectate lyase, and Chorismate mutase were chosen for molecular docking studies from the guava root-knot nematode, *Meloidogyne enterolobii* were screened and selected (Menezes *et al.,* 2019, Santos *et al.,* 2019, Mani *et al.,* 2021, FENG *et al.,* 2023, Zahoor *et al.,* 2023). The selected five target proteins were screened based on previously published reports highlighting their essential roles in nematode biology and pathogenicity. An earlier study has identified cellulose-binding proteins as key mediators of host cell wall degradation and root penetration in root-knot nematodes. Cytochrome oxidase subunit I has been widely reported as a critical component of mitochondrial respiration and energy metabolism, essential for nematode survival and infectivity. Similarly, other selected proteins have been documented in the literature for their involvement in neuromuscular regulation, stress response, and developmental processes, and functionally validated targets in plant-parasitic nematodes. Though this is the first study on selected protein models of *M. enterolobii*, it lacks both experimentally and computationally determined structures. Hence, the molecular modelling was performed using the I-TASSER tool (Zhou *et al.,* 2022). Homology modelling is the computational approach used to develop the three-dimensional structure of the target proteins. The FASTA sequence of the required protein is submitted to I-TASSER. The results were generated with five models and one optimum protein model was selected. The cellulose-binding protein had a C-Score (1.67) which has an estimated TM-score of 0.95±0.05, and an estimated RMSD value of 1.7±1.5. The cytochrome oxidase subunit I protein had a C-Score (1.45) which has an estimated TM-score of 0.85±0.05, and an estimated RMSD value of 1.5±1.5. The major sperm protein had a C-Score (1.54) which has an estimated TM-score of 0.82±0.05, and an estimated RMSD value of 1.8±1.5. The chorismate mutase protein had a C-Score (1.63) which has an estimated TM-score

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of 0.92±0.05, and an estimated RMSD value of 1.5±1.5 and the pectate lyase protein had a C-Score (1.68) which has an estimated TM-score of 0.91±0.05, and an estimated RMSD value of 1.5±1.5.

2.9. Validation of developed protein model and ligand preparation

The modelled proteins, Cellulose binding protein, Major sperm protein, Cytochrome oxidase subunit I, Pectate lyase, and Chorismate mutase were validated by generating a Ramachandran plot using the PROCHECK tool (LASKOWSKI *et al.,* 2013). The stereochemical integrity of the models was evaluated through the Ramachandran plot, performed via the PROCHECK module available on the Structural Analysis and Verification Server (SAVES) (https://saves.mbi.ucla.edu/). The modelled proteins were analyzed for residues located in the favored and allowed regions of the Ramachandran plot. The residues outside the favoured region were corrected using loop building by modeler 10.4.

2.10. Preparation of ligands and molecular docking assays

The biomolecules identified from effective *Streptomyces* spp. strains through GC-MS analysis results were selected for further molecular docking studies. The three-dimensional structures of the chosen bioactive compounds were obtained from the PubChem database (https://pubchem.ncbi.nlm.nih.gov/). The nematicidal compound carbofuran 3G (PubChem ID: 2566) was used as a control for comparative analysis. Molecular docking of the identified biomolecules was conducted using the Vina module within PyRx 0.8 software. Proteins were prepared by converting them into macromolecules using the "make macromolecule" function in PyRx 0.8. The three-dimensional structure of the modelled protein targets of *M. enterolobii* was docked with effective ligand molecules to study their intermolecular interactions. The ligands from *Streptomyces* spp. strains were energy-minimized using the Universal Force field (UFF). During protein preparation, polar hydrogens were added, and partial atomic charges were assigned. The gasteiger charges were also applied, non-polar hydrogens were merged and rotatable bonds were identified based on the ligand structures. Protein-ligand interactions were analyzed and visualized using Biovia Discovery Studio Visualizer 24.1.0.23298. The docking results were interpreted based on binding energies, with lower binding energies indicating stronger affinities. Potential binding sites were predicted using the CastP (Computed Atlas of Surface Topography tool).

2.11. Pot experiment

Six-month-old guava saplings (25cm tall) of the cv. Arka Kiran was planted in 4kg pots (15× 18.5 × 21 cm) filled with a sterilized potting mixture of clay soil, and farmyard manure (2:1:1 ratio). Each treatment consisted of five replications with each replicate comprising three guava saplings (n=15 per treatment). Additionally, carbofuran @1 g/plant was included as a standard check. Soil drenching done with culture filtrates of *Streptomyces* spp. on guava (*Psidium guajava*) plants inoculated with *M.*

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enterolobii. Three *Streptomyces* spp. strains that exhibited superior antagonistic activity in vitro studies, i) *Streptomyces sparsus* GHRS-2, ii) *Streptomyces* sp. strain GHS-3, iii) *Streptomyces* sp. strain GHRS-5 were selected for conducting pot experiments. Cell-free culture suspensions containing 10^4 colony forming units/ml of each *Streptomyces* strain were prepared as detailed earlier. Soil drenching was performed using the whole culture filtrate at a concentration of 30ml/l, while distilled water was used as the untreated control. Half a liter of the treatment suspension was drenched into the soil immediately after planting, following by inoculation with 5000 *M. enterolobii* J_2 juveniles per pot. The experiment was maintained at 50% shade and a temperature of 26 ± 2°C. plant biometric characters, including shoot and root lengths and fresh shoot and root weights, were measured. Nematode infection parameters such as J_2 population in soil, female population in roots, egg mass counts, and root gall indices were also assessed. Root gall indices were evaluated on a 1-10 rating scale after carefully washing the roots to remove adhering soil. Potting mixture samples (250ml per pot) were thoroughly mixed and processed, and the female nematode population and egg mass counts per gram of root were recorded from dissected roots. The entire glasshouse experiment, including treatments, was repeated once under identical conditions.

2.12. Statistical analysis

The observed data were subjected to statistical analysis using R software (2023.09.1+494), and one-way analysis of variance (ANOVA) and mean values were compared by Duncan's multiple range test.

3. Results

3.1. Identification of nematode species

The study focused on identifying a nematode affecting guava through detailed morphological and molecular characterization. Morphological examination of the posterior cuticular pattern revealed oval-shaped dorsal arches with a combination of squarish and rounded features, displaying both smooth and coarse striations. Molecular analysis targeted the amplification of the 16S rRNA gene region across various life stages of *M. enterolobii* producing 250bp fragments. The BLAST searches showed 99% similarity with reference sequences of *M. enterolobii*. The aligned DNA sequences were submitted to GenBank, and an accession number was assigned [PQ098052].

3.2. *In vitro* evaluation of nematicidal activity

Among the 15 *Streptomyce*s spp. isolates, GHRS-5, GHRS-2, GHS-3, and GHRS-4 showed effectiveness in preliminary assays against *M. enterolobii*. GHRS-5 was the most potent, achieving over 95% reduction compared to the control (Distilled H_2O). After 72 hours, GHRS-5-treated eggs had the lowest hatching rate (32.6 hatched J_2), significantly lower than the control (310.9 hatched J_2). Filtrates of

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GHRS-2 (78.5 hatched J_2) and GHS-3 (58.9 hatched J_2) also inhibited egg hatching (Table 1) (Fig 2). Juvenile mortality increased over the incubation period, with GHRS-5 resulting in the highest mortality (100 dead J_2), followed by GHS-3 (90.2 dead J_2), and GHRS-2 (88.8 dead J_2) (Table 2) (Fig 3). Only minimal juvenile mortality (10.1 dead J_2) was observed in the control, likely due to natural causes.

3.3. Identification of effective *Streptomyces* spp. isolates

Among the various *Streptomyces* spp. strains tested, GHRS-2, GHS-3, and GHRS-5 demonstrated significant nematicidal activity and were selected for further characterization. The 16S rRNA gene of these strains was amplified and sequenced using Sanger sequencing, and the sequences were submitted to the NCBI database. The identified *Streptomyces* spp. strains and their corresponding NCBI accession numbers are as follows: *Streptomyces sparsus* GHRS-2 (PQ098148), *Streptomyces* sp. strain GHS- 3 (PQ098124), *Streptomyces* sp. strain GHRS- 5 (PQ098144).

3.4. Profiling the effective biomolecules of *Streptomyces* spp. GHRS-5

The biomolecules of *Streptomyces* spp. GHRS-5 were analyzed using the GC-MS analysis. The results also revealed that GHRS-5 colonies produced a diverse range of unique secondary metabolites, potentially playing a crucial role in nematode suppression. The GC-MS results revealed the presence of 9(11)-Dehydroergosterol tosylate, (5*aR*)-5,5-difluoro-1-[3-[2-[(3*R*)-3-hydroxy-1-methyl-2-oxopyrrolidin-3-yl] ethynyl] phenyl]-5*a*-methyl-4*a*,6-dihydro-4*H*-cyclopropa[f]indazole-3-carboxamide, 9,12,15-Octadecatrienoic acid, 2-phenyl-1,3-dioxan-5-yl ester, Estra-1,3,5(10)-trien-17á-ol, 5alpha-Pregnane-3beta,11beta,20beta,21-tetrol, 17-Methoxyandrost-4-en-3-one o-methyloxime. Also, the pathway analysis was used to interpret the significance of the identified pathways with compounds from GHRS-5 (Fig 4) interconnection pathway being shared among the effective biomolecules. Also, compound reaction metabolism has been analysed from effective biomolecules using metscap in cytoscape 3.10.1 (Fig 5). Furthermore, the TCA cycle and associated metabolic network highlight the interaction between compounds, genes, reactions, and enzymes linked to bioactive metabolites produced by *Streptomyces* spp. GHRS-5 (Fig 6). This integrative representation aids in understanding the metabolic pathways driving its nematicidal activity. Additionally, a correlation network diagram was constructed for *Streptomyces* spp. GHRS-5 which illustrates the interconnections among its effective biomolecules and their respective functional groups. Nodes represent biomolecules categorized by their standard groups (eg., lipopeptides, polyketides, and volatile organic compounds), while edges signify relationships such as co-expression, biosynthetic pathway links, and shared targets in nematode suppression. This visualization highlights functional clustering and the biomolecule's synergistic roles (Fig 7).

3.5. Molecular modelling and validation of protein targets

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The modelled protein targets were visualized using BIOVIA Discovery Studio v21.1.0.20298 (Fig 8) and assessed for residue distribution in allowed, additionally allowed, and disallowed regions using the Ramachandran plot. Also, the target protein was evaluated for residues in the most favored regions and generously allowed regions (Supplementary figure S1).

3.6. Virtual screening and molecular docking of target proteins with identified biomolecules

All the bioactive molecules from GHRS-5 exhibited a significant impact on various protein targets of *M. enterolobii.* Among the effective biomolecules screened, 9(11)-Dehydroergosterol tosylate and (5*aR*)-5,5-difluoro-1-[3-[2-[(3*R*)-3-hydroxy-1-methyl-2-oxopyrrolidin-3-yl] ethynyl] phenyl]-5*a*-methyl-4*a*,6-dihydro-4*H*-cyclopropa[f]indazole-3-carboxamide showed the most prominent effect with highest binding affinity and lowest energy values (Table 3). These effective biomolecules showed dominant results in all the selected proteins as cellulose-binding protein displayed the highest binding affinity with (-10.4 kcal/mol & -9.2 kcal/mol) (Fig 9), followed by cytochrome oxidase subunit I showed the highest binding affinity with (-10.4 kcal/mol & -8.7 kcal/mol) (Fig 10). Similarly, major sperm protein showed the highest binding affinity with (-9.9 kcal/mol & -9.1 kcal/mol) (Fig 11). Also, chorismate, mutase shows the highest binding affinity with (-8.8 kcal/mol & -8.1 kcal/mol) (Fig 12). Furthermore, pectate lyase showed the highest binding affinity with (-7.2 kcal/mol & -9.0 kcal/mol) (Fig 13) (Table 4).

3.7. Pot experiment

The three *Streptomyces* spp. isolates that performed well under in vitro conditions were taken for pot studies for further assessment of their efficacy on *M. enterolobii* population and plant growth parametric characters. Also, the seedling dip treatment of all the *Streptomyces* spp. isolates had significant effect on nematode population and plant growth parameters. There was significant difference amongst *Streptomyces* spp. isolates in both plant growth and nematode reproductive parameters. GHRS-5 showed the highest effect on promoting the plant growth parameters which exhibited around an 82% increase over control on root length and a 60% increase in shoot length (Table 5; Fig 14). It demonstrated significant potential in suppressing the nematode population in roots with a 79.2% reduction in egg masses per root system compared to the control. Additionally, reduction of 62% in female nematodes and 65% in the soil population of *Meloidogyne enterolobii* were observed over the control (Table 6; Fig 15).

4. 4. Discussion

The findings from this study highlight the growing threat of PPN, particularly *M. enterolobii* to global agriculture and the immediate need for sustainable control measures. *M. enterolobii* known for

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its broad host range adaptability and high reproductive rate poses a severe challenge to conventional agricultural practices due to its ability to overcome plant-resistance genes and proliferate in resistant crops. This nematode infestation of guava, a major crop region such as India, results in significant yield losses and economic setbacks for farmers, emphasizing the demand for innovative eco-friendly pest control strategies. Chemical nematicides are though effective have faced increasing scrutiny due to their environmental and health hazards. The use of biological control agents, particularly *Streptomyces* spp. as a sustainable and eco-friendly approach for managing PPNs has gained increasing attention in agricultural research. This novel study focuses on the potential of *Streptomyces* spp. metabolites in controlling *M. enterolobii*, a highly aggressive root-knot nematode. The findings align with previous studies on the role of microbial secondary metabolites in combating PPN and contribute to the growing body of knowledge on nematode management through natural compounds. Several studies have demonstrated the nematicidal potential of secondary metabolites from microbial sources supporting the current research findings. For instance, bacterial genera such as *Bacillus, Pseudomonas*, and *Streptomyces* are widely reported to produce bioactive compounds with nematicidal properties (Fischer *et al.,* 2013).

In this study, we explored *Streptomyces* spp. soil-borne bacteria known for their rich production of secondary metabolites with potential nematicidal properties. The effectiveness of *Streptomyces* spp. strains as nematicidal agents against *M. enterolobii* were demonstrated through *in vitro* assays on egg hatching and juvenile mortality. Among the isolates tested, strain GHRS-5 exhibited remarkable antagonistic effects on both nematode eggs and juveniles. The bioactive compounds produced by this strain analyzed via GC-MS revealed a diverse array of nematicidal biomolecules capable of significantly reducing nematode populations. These findings align with the previous studies that underscore the role of bacterial metabolites such as antibiotics, enzymes, and lipopeptides in combating PPN (Zheng *et al.,* 2018). Similarly, the current study identifies nematicidal activity in metabolites produced by *Streptomyces* spp., particularly strain GHRS-5, which exhibited the highest efficacy in reducing egg hatching and juvenile mortality of *M. enterolobii.* Studies conducted by (Proença *et al.,* 2019) explored the nematicidal effects of bacterial metabolites on PPNs, focusing on compounds like antibiotics, volatile organic compounds (VOCs) and enzymes. The findings from this study are consistent with their conclusions as *Streptomyces* strains in the present study were found to produce metabolites with significant inhibitory effects on nematode egg hatching and juvenile mortality. Notably, metabolites such as volatile organic compounds (eg., geosmin) and antimicrobial enzymes have been identified in *Streptomyces* spp. as potent bioactive agents which aligns with the GC-MS profiling conducted in the current study.

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A recent study by (Park *et al.,* 2023) demonstrated the nematicidal potential of *Streptomyces* strains through the production of cellulases and chitinases that degrade nematode eggs and disrupt juvenile cuticles. The current study findings with these results, as the metabolic profiling revealed enzymes and VOCs produced by GHRS-5 that directly target nematode structures. These similarities underline the universal applicability of *Streptomyces* spp. in biocontrol mechanisms and highlight the importance of metabolite profiling for identifying key compounds responsible for nematicidal activity.

The integration of molecular docking techniques to identify potential protein targets of *M. enterolobii* provides an innovative perspective in this study, enabling precise evaluation of biomolecule efficacy. Similar approaches have been reported by (Shanmugam *et al.,* 2018) who utilized *in silico* methods to study interactions between nematicidal compounds and nematode proteins. Their research emphasizes the potential of molecular modelling to optimize biocontrol strategies by applying the approach adopted in this study to screen bioactive compounds from *Streptomyces* isolates. In comparison with chemical nematicides such as oxamyl and fluensulfone, the eco-friendly and sustainable nature of microbial metabolites becomes evident. (Chen *et al.,* 2020) noted that synthetic nematicides face significant limitations due to environmental concerns, high costs, and declining efficacy over time. The molecular docking results provide supportive evidence for potential interactions between the identified biomolecules and key nematode proteins. However, these findings should be considered preliminary and hypothesis-generating in nature. Computational binding affinity does not inherently confirm functional inhibition under biological conditions. The actual nematicidal efficacy may be influenced by additional factors, including compound diffusion across the nematode cuticle, metabolic transformation, protein conformational dynamics, and *in vivo* target accessibility. Therefore, the proposed mechanisms of nematode suppression derived from docking analysis require further experimental validation. Future studies involving *in vitro* enzyme inhibition assays, target-specific gene silencing (RNAi), or protein expression analyses would be necessary to substantiate the predicted molecular interactions. Furthermore, (Gopalakrishnan *et al.,* 2020) highlighted the role of *Streptomyces* as both biocontrol agents and plant growth promoters. The dual functionality observed in the present study where GHRS-5 suppressed *M. enterolobii* and also underscores its potential for field application.

The study underscores the effectiveness of *Streptomyces* spp. isolates in reducing *Meloidogyne enterolobii* populations and improving plant growth under pot conditions. Among the isolates, GHRS-5 showed the highest efficacy, promoting root and shoot length by 82% and 60% respectively compared to the control. Additionally, GHRS-5 demonstrated significant nematicidal activity by reducing egg masses per root system by 79.2% female nematode populations by 62%, and soil nematode populations by 65%. These findings align with earlier reports highlighting the role of *Streptomyces* spp. in suppressing root-knot nematodes and promoting plant growth (Hyder *et al.,* 2021). The variations

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observed among isolates suggest that the efficacy of *Streptomyces* strains depends on their unique bioactive compounds and interactions with plant and soil environments. The results highlight the potential of GHRS-5 as a biocontrol agent and plant growth promoter, paving the way for its application in sustainable nematode management strategies. Although such growth enhancement and nematode suppression could be associated with activation of host defense responses, the present experimental design did not include split-root systems or molecular and biochemical analyses required to confirm induced systemic resistance (ISR). Therefore, ISR cannot be conclusively established in this study and remains a potential mechanism that warrants further practical implementation.

The mechanism by which *Streptomyces* spp. act against nematodes appears multifaceted, involving direct nematicidal activity and possible induction of systemic resistance in host plants. Through molecular docking studies, this study elucidated the binding affinities of *Streptomyces* metabolites with specific nematode protein targets providing insights into how these compounds may interfere with nematode physiology and survival. The ability of *Streptomyces* metabolites to interact with proteins essential for nematode development, such as cellulose-binding protein and chorismate mutase suggests a targeted mode of action that could reduce nematode reproduction and establishment in plants. Our results demonstrate the potential of using *Streptomyces* spp. as a bionematicide against *M. enterolobii* under both controlled and field conditions. Moreover, the current study's findings align with those of (Al-Tammar and Khalifa 2023) who demonstrated the effectiveness of rhizosphere derived *Streptomyces* isolates in suppressing various PPN species. The isolation of GHRS-5 from guava rhizosphere soils in this study not only validates the potential of local microbial communities for nematode management but also emphasizes the importance of exploring indigenous microbial resources for sustainable agriculture. By focusing on naturally derived microbial metabolites, this study contributes to sustainable agricultural practices, offering an alternative to chemical nematicides that not only protect crop yields but also promote environmental and soil health. The identified effective biomolecules may be further investigated with novel formulation sources and *in silico* analysis. Continued exploration of *Streptomyces* spp. and other biocontrol agents will be crucial in developing integrated pest management (IPM) strategies that address the growing issue of nematode resistance and safeguard global food security.

5. Conclusions

This study highlights the growing threat posed by *M. enterolobii* to global agriculture and underscores the immediate need for sustainable nematode management strategies. The high adaptability, reproductive potential, and ability to overcome plant resistance make this invasive nematode particularly challenging for conventional control methods. The findings demonstrate the potential of *Streptomyces*

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spp. GHRS-5 is an effective and eco-friendly biocontrol agent. Through metabolic profiling and in vitro assays, strain GHRS-5 exhibited significant nematicidal activity, reducing egg hatching and promoting juvenile mortality. The current research reinforces the multifaceted action of *Streptomyces* metabolites including molecular docking analysis provided deeper insights into the targeted interactions between microbial metabolites and nematode proteins supporting the development of specific efficient bionematicides. Exploring indigenous microbial resources like GHRS-5 opens opportunities for sustainable agriculture, protecting crop yields while improving soil health. Future studies should focus on optimizing field applications and developing novel formulations to maximize the efficacy of these promising biocontrol agents.

CRediT authorship contribution statement

Janani Mani: Writing – original draft, Formal analysis, Methodology, Data curation, software, review & editing. **Seenivasan Nagachandrabose:** Conceptualization, Project administration, Supervision, review & editing. **R. Thamizh Vendan:** Supervision, Resources, visualization **Jayakanthan Mannu:** Formal analysis, software, Validation, **Prabhu Somasundaram:** Conceptualization, Visualization. **Devrajan Kandasamy:** Supervision, Resources, visualization. **S. Harish:** Supervision, visualization. **Rajkumari Jyotika:** Software, **Ragasruthi Malaravan:** Software.

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Treatments	**Number of eggs hatched ***

Concentrations	25%				50%				75%				100%			
	24h	48h	72h	96h	24h	48h	72h	96h	24h	48h	72h	96h	24h	48h	72h	96h
Streptomyces sparsus GHRS-2	49.7[c] (7.04)	73.1[c] (8.54)	86.5[c] (9.30)	97.8[c] (9.88)	55.8[c] (7.46)	72.1[c] (8.49)	88.9[c] (9.42)	110.8[c] (10.52)	55.9[c] (7.47)	68.4[c] (8.27)	89.2[c] (9.44)	98.1[c] (9.90)	15.9[c] (3.98)	45.9[c] (6.77)	48.8[c] (6.98)	78.5[c] (8.86)
Streptomyces spp. GHS-3	40[b] (6.32)	66.2[b] (8.14)	72.3[b] (8.50)	89.9[b] (9.48)	40.2[b] (6.34)	60.3[b] (7.77)	78.3[b] (8.85)	92.1[b] (9.60)	45.8[b] (6.77)	55.4[b] (7.44)	72[b] (8.49)	87.4[b] (9.35)	10.2[b] (3.19)	30.2[b] (5.50)	35.9[b] (5.99)	58.9[b] (7.67)
Streptomyces spp. GHRS-5	30.2[a] (5.50)	58.1[a] (7.62)	62.0[a] (7.87)	80.1[a] (8.95)	38.2[a] (6.18)	61.8[a] (7.86)	72.3[a] (8.50)	86.4[a] (9.30)	35.5[a] (5.96)	50.5[a] (7.11)	66.1[a] (8.13)	70.3[a] (8.38)	3.4[a] (1.84)	25.1[a] (5.01)	28.0[a] (5.29)	32.6[a] (5.71)
Nutrient broth	76[cd] (8.72)	110[de] (10.49)	258[cd] (16.06)	380[de] (19.49)	72[cd] (8.49)	108[de] (10.39)	230[cd] (15.17)	330[ef] (18.17)	90[c] (9.49)	98.9[c] (9.94)	180.9[de] (13.45)	220.8[cd] (14.86)	20[c] (4.47)	80.5[cd] (8.97)	160.2[de] (12.66)	254.2[ef] (15.94)
Control	97.2[ef] (9.85)	150.2[fg] (12.26)	220.7[g] (14.86)	320.6[gh] (17.91)	80.5[de] (8.98)	140.2[fg] (11.84)	220.8[gh] (14.86)	350.2[hi] (18.71)	80.2[ef] (8.95)	140.4[fg] (11.85)	282.5[gh] (16.80)	350.5[hi] (18.72)	76.8[g] (8.76)	125.3[h] (11.19)	290.2[hi] (17.03)	310.9[hi] (17.63)
Sed	1.26	1.76	1.63	1.97	1.78	1.58	1.87	1.54	1.66	1.89	1.90	1.33	1.67	1.03	1.87	1.92
CD (0.05)	2.52	2.02	2.49	2.59	3.13	2.98	2.98	2.68	2.21	2.12	2.34	2.49	2.56	2.12	2.14	2.43

Table 1. Effect of different concentration of culture filtrate of *Streptomyces* spp. strains *M. enterolobii* egg hatching

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* Data represent pooled means of two independent experimental runs. Preliminary analysis showed no significant effect of experimental run ($p > 0.05$); therefore, data were combined and analyzed using one-way ANOVA. Means followed by different letters differ significantly at $p \leq 0.05$. Values in parenthesis represents square root transformed values. Analysis of variance was performed through R software (Version: 2023.09.1+494).

Treatments	**Number of juveniles dead***															
Concentrations	**25%**				**50%**				**75%**				**100%**			
	24h	**48h**	**72h**	**96h**	**24h**	**48h**	**72h**	**96h**	**24h**	**48h**	**72h**	**96h**	**24h**	**48h**	**72h**	**96h**
Streptomyces spp. GHRS-2	15.8^{b} (3.97)	18.2^{bc} (4.27)	33.5^{bc} (5.78)	45.8^{c} (6.77)	28.2^{c} (5.31)	45.9^{c} (6.77)	55.2^{c} (7.42)	60.1^{cd} (7.75)	32.1^{c} (5.66)	48.6^{c} (6.97)	60.8^{d} (7.79)	72.3^{de} (8.50)	43.2^{d} (6.57)	72.8^{de} (8.53)	78.4^{de} (8.85)	88.8^{ef} (9.42)
Streptomyces spp. GHS-3	19.7^{ab} (4.43)	22.3^{ab} (4.72)	45.8^{b} (6.77)	55.8^{b} (7.47)	32.1^{ab} (5.67)	50.3^{ab} (7.09)	60.8^{b} (7.79)	65.3^{b} (8.08)	45.2^{b} (6.72)	55.8^{a} (7.47)	80.4^{b} (8.97)	88.4^{a} (9.40)	55.2^{b} (7.42)	88.9^{ab} (9.42)	89.1^{b} (9.43)	90.2^{ab} (9.49)
Streptomyces spp. GHRS-5	28^{a} (5.29)	35.8^{a} (5.98)	50.8^{a} (7.13)	62.3^{a} (7.90)	38.2^{a} (6.18)	55.8^{a} (7.46)	62.9^{a} (7.93)	72.5^{a} (8.51)	52.4^{a} (7.24)	74.2^{a} (8.61)	88.5^{ab} (9.40)	91.1^{a} (9.54)	60.5^{b} (7.77)	90.22^{a} (9.49)	100^{a} (10)	100^{a} (10)
Nutrient broth	0^{d} (0)	6.5^{d} (2.54)	12.43^{d} (3.56)	14.5^{de} (3.81)	0^{d} (0)	8.52^{de} (2.91)	10.75^{de} (3.27)	12.45^{ef} (3.52)	0^{d} (0)	7.5^{de} (2.73)	12.5^{ef} (3.53)	15.3^{ef} (3.91)	0^{d} (0)	7.35^{d} (2.71)	10.5^{d} (3.24)	20.45^{de} (4.52)
Control	0^{e} (0)	3.2^{de}	8.2^{de}	9.1^{de}	0^{e} (0)	3.2^{de}	8.45^{ef}	10.5^{ef}	0^{e} (0)	4.22^{de}	9.12^{e}	10.24^{e}	0^{e} (0)	5.23^{ef}	8.1^{f}	10.1^{f}

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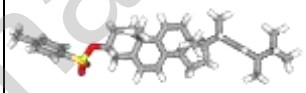
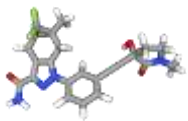
		(1.79)	(2.87)	(3.01)		(1.78)	(2.90)	(3.24)		(2.05)	(3.01)	(3.19)		(2.28)	(2.84)	(3.17)
CD (0.05)	1.03	1.23	1.34	1.45	1.23	1.14	1.17	1.21	1.34	1.42	1.56	1.72	1.02	1.14	1.21	1.31
Sed	2.12	2.42	2.56	2.61	2.42	2.51	2.67	2.78	2.83	2.55	2.78	2.90	2.14	2.23	2.45	2.72

Table 2. Effect of different concentrations of culture filtrate of *Streptomyces* spp. strains against *M. enterolobii* on juvenile mortality

*Data represent pooled means of two independent experimental runs. Preliminary analysis showed no significant effect of experimental run ($p > 0.05$); therefore, data were combined and analyzed using one-way ANOVA. Means followed by different letters differ significantly at $p \leq 0.05$. Values in parenthesis represents square root transformed values. Analysis of variance was performed through R software (Version: 2023.09.1+494).

Table 3. Binding affinity values for nematicidal metabolites from *Streptomyces* sp. GHRS-5 against different virulent protein targets of *M. enterolobii*

				Binding affinity (kcal/mol) of biomolecules against target proteins of *M. enterolobii*				
Sl. No.	**Biomolecules**	**Pubchem compound CID**	**Molecular Structure**	**Cellulose Binding protein**	**Cytochrome oxidase subunit I**	**Major sperm protein**	**Chorismate mutase**	**Pectate lyase**
1	9(11)-Dehydroergosterol tosylate	Cid 91691728		-10.4	-10.4	-9.9	-8.8	-7.2
2	(5*aR*)-5,5-difluoro-1-[3-[2-[(3*R*)-3-hydroxy-1-methyl-2-oxopyrrolidin-3-yl] ethynyl] phenyl]-5*a*-methyl-4*a*,6-dihydro-4*H*-cyclopropa[f]indazole-3-carboxamide	Cid 117923523		-9.2	-8.7	-9.1	-8.1	-9.0
3	9,12,15-Octadecatrienoic acid, 2-phenyl-1,3-dioxan-5-yl ester	Cid 556671	(5*aR*)-5,5-difluoro-1-[3-[2-[(3*R*)-3-hydroxy-1-methyl-	-8.4	-6.8	-8.0	-8.1	-6.2

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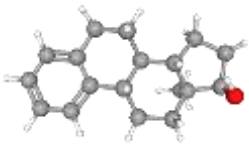
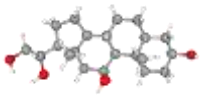
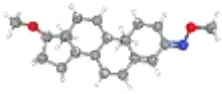
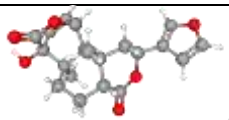
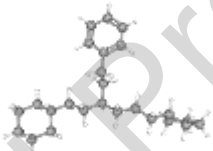
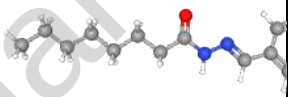
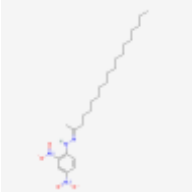
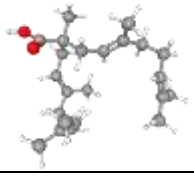
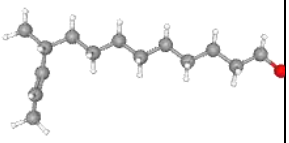
			2-oxopyrrolidin-3-yl] ethynyl] phenyl]-5*a*-methyl-4*a*,6-dihydro-4*H*-cyclopropa[f]indazole-3-carboxamide					
4	Estra-1,3,5(10)-trien-17á-ol	Cid 13058341		-8.3	-7.4	-8.0	-8.3	-6.8
6	5alpha-Pregnane-3beta,11beta,20beta,21-tetrol	Cid 192841		-8.3	-7.4	-7.5	-7.8	-6.6
7	17-Methoxyandrost-4-en-3-one o-methyloxime	Cid 91712453		-8.1	-8.1	-8.5	-8.2	-6.7
8	Columbin	Cid 442015		-8.1	-8.6	-8.9	-7.4	-7.6
9	Benzene, [3-(2-cyclohexylethyl)-6-cyclopentylhexyl]-	Cid 279536		-8.1	-5.8	-7.0	-6.8	-6.0
10	N'-(4-Nitrobenzylidene) octanohydrazide	Cid 6187499		-7.9	-6.4	-7.0	-6.1	-5.9
11	2-Nonadecanone 2,4-dinitrophenylhydrazine	Cid 9601657		-7.1	-6.4	-6.5	-7.2	-5.3
12	2,6,9,12,16-Pentamethylheptadeca-2,6,11,15-tetraene-9-carboxylic acid	Cid 5365892		-6.6	-6.2	-7.7	-7.0	-5.9
13	10-Methyl-E-11-tridecen-1-ol propionate	Cid 5365029		-5.1	-4.4	-5.1	-4.9	-4.6

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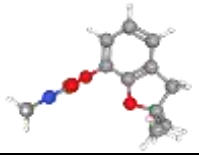
15	Carbofuran	Cid 2566		-5.5	-5.9	-6.1	-5.2	-5.2

Table 4. Molecular docking studies on different protein targets of *Meloidogyne enterolobii* with nematicidal biomolecules

Target protein of *M. enterolobii*	**Name of a ligand**	**Binding affinity (kcal/mol)**	**Residues involved in hydrogen bond formation**	**Residues involved in Vander Waals attraction**
Cellulose binding protein	9(11)-Dehydroergosterol tosylate	-10.4	LYS 53	VAL 114, PHE 82, GLY 86, THR 50, LEU 87, ILE 46, LEU 90, ILE 83, ALA 47, VAL 54, GLY 51, PHE 55, THR 43, GLY 89, HIS 24, LEU 92, PHE 111
	Carbofuran	-5.5	VAL 155, ASN 156	SER 116, VAL 108, PRO 161, ALA 109, PHE 160, MET 157,
Cytochrome	(5*aR*)-5,5-difluoro-1-[3-[2-[(3*R*)-3-	-8.7	ASN 92	LEU 114,

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selected protein	hydroxy-1-methyl-2-oxopyrrolidin-3-yl] ethynyl] phenyl]-5*a*-methyl-4*a*,6-dihydro-4*H*-cyclopropa[f]indazole-3-carboxamide			PRO 192, TYR 208, ILE 69, GLY 87, SER 90, SER 11,ASN 91
	Carbofuran	-5.9	ARG 166	VAL 112, THR 144, PHE 141, PHE 128
Major sperm protein	(5*aR*)-5,5-difluoro-1-[3-[2-[(3*R*)-3-hydroxy-1-methyl-2-oxopyrrolidin-3-yl] ethynyl] phenyl]-5*a*-methyl-4*a*,6-dihydro-4*H*-cyclopropa[f]indazole-3-carboxamide	-9.1	GLY 332, ALA 94	SER 189, THR 91, GLY 333, PRO 290, PHE 292, VAL 291, PRO 100, PRO 89, LYS 339, ARG 200
	Carbofuran	-6.1	-	VAL 110, ARG 210, PRO 85
Chorismate mutase	9(11)-Dehydroergosterol tosylate	-8.8	ARG 54	LEU 92, PHE 47, LEU 94, LEU 81
	Carbofuran	-5.2	ASP 97	THR 100, ASN 95, VAL 99, LEU 101

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Pectate lyase	2-Nonadecanone 2,4-dinitrophenylhydrazine	-5.3	ASN 94	GLU 117, CYS 65, CYS 8, GLU 9, ILE 45, ASP 11, ASN 69, VAL 93, THR 40
	Carbofuran	-5.2	TYR 138	PHE 108, GLY 84, ARG 60, GLY 110, GLN 137, GLN 107, THR 81

Table 5. Influence of *Streptomyces* sp. strain GHRS-5 on plant growth biometric parameters under greenhouse conditions.

Treatment N.	Treatment	Root length	Shoot length	Fresh weight	Shoot weight	Root weight
1	*Streptomyces* spp. GHRS-2	18.2[c] (4.26)	16.2[c] (4.02)	10.1[c] (3.17)	12.2[c] (3.49)	11.9[c] (3.45)
2	*Streptomyces* spp. GHS-3	21.5[b] (4.63)	22.6[b] (4.75)	9.80[b] (3.13)	18.3[b] (4.27)	8.5[b] (2.91)
3	*Streptomyces* spp. GHRS-5	43.2[a] (6.57)	25.5[a] (5.04)	13.2[a] (3.63)	25.3[a] (5.02)	13.2[a] (3.63)
4	Chemical check	15.2[d] (3.89)	16[d] (4)	8[d] (2.82)	10.8[d] (3.28)	6.7[d] (2.58)
5	Control	11[de] (3.31)	12.3[d] (3.50)	5.6[d] (2.36)	7[e] (2.64)	5.4[e] (2.32)
	SE	0.52	0.87	1.18	1.54	1.52
	CD (0.05)	1.09	1.76	1.32	2.01	2.32

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* Data represent pooled means of two independent experimental runs. Preliminary analysis showed no significant effect of experimental run ($p > 0.05$); therefore, data were combined and analyzed using one-way ANOVA. Means followed by different letters differ significantly at $p \leq 0.05$. Values in parentheses show % increase (+) over control.

Table 6. Effect of *Streptomyces* sp. strain GHRS-5 on nematode reproduction parameters under greenhouse conditions.

Treatment No.	Treatment	Females / g root*	Egg masses / g root	Gall Index	J_2 population (in 250cc soil)
1	*Streptomyces* spp. GHRS-2	32.2c (5.67)	17.3^{b} (4.15)	3	94.2^{b} (9.70)
2	*Streptomyces* spp. GHS-3	25.2^{b} (5.01)	10.2^{b} (3.19)	2	92.8^{b} (9.63)
3	*Streptomyces* spp. GHRS-5	20^{a} (4.47)	12.2^{a} (3.49)	2	85.2^{a} (9.23)
4	Chemical check	39.1^{cd} (6.25)	23.2^{d} (4.81)	4	102.1^{cd} (10.10)
5	Control	44.7^{de} (6.68)	26.8^{e} (5.17)	4	123.5^{ef} (11.11)
	SE	1.08	1.45	-	2.97
	CD (0.05)	2.21	1.20	-	4.59

* Data represent pooled means of two independent experimental runs. Preliminary analysis showed no significant effect of experimental run ($p > 0.05$); therefore, data were combined and analyzed using one-way ANOVA. Means followed by different letters differ significantly at $p \leq 0.05$. Values in parentheses show a % decrease (-) over control.

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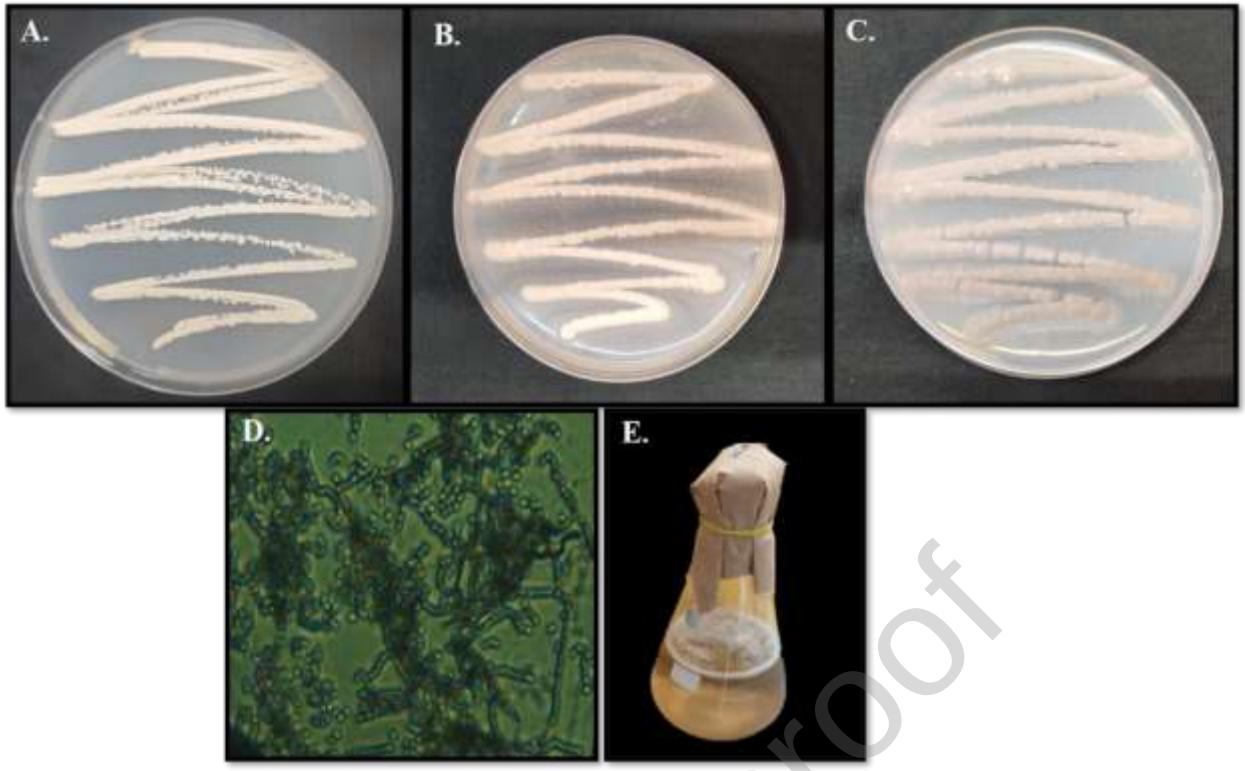

Figure 1. Cultivation characteristics of (A) *Streptomyces sparsus* GHRS-2, (B) *Streptomyces* sp. GHS-3, (C) *Streptomyces* sp.

GHRS-5 (D)Aerial hyphae morphology of *Streptomyces* sp. GHRS-5 (E) Ken knight's medium preparation

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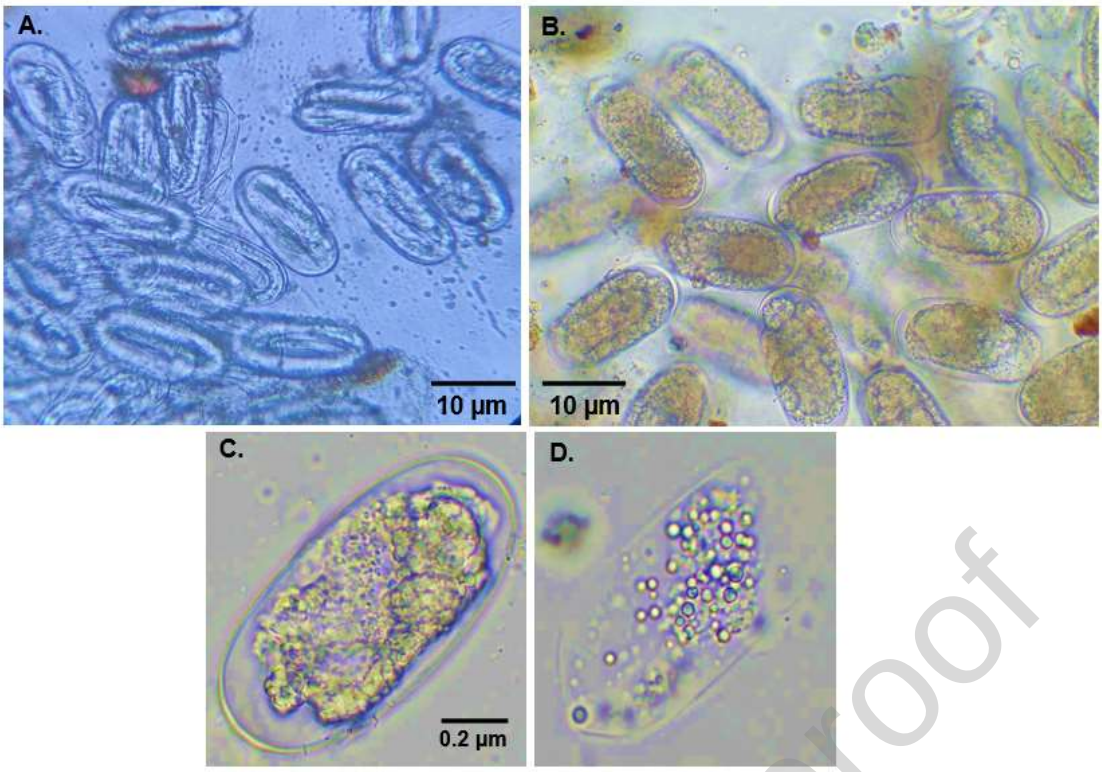

Figure 2. Effect of *Streptomyces* sp. GHRS-5 culture filtrate on *M. enterolobii* egg parasitization

(A) Healthy egg mass of *M. enterolobii* (B) Infected egg mass with spores of *Streptomyces* sp. GHRS-5
(C) *Streptomyces* sp. GHRS-5 spore-infected egg (D) *Streptomyces* sp. GHRS-5 vegetative spore

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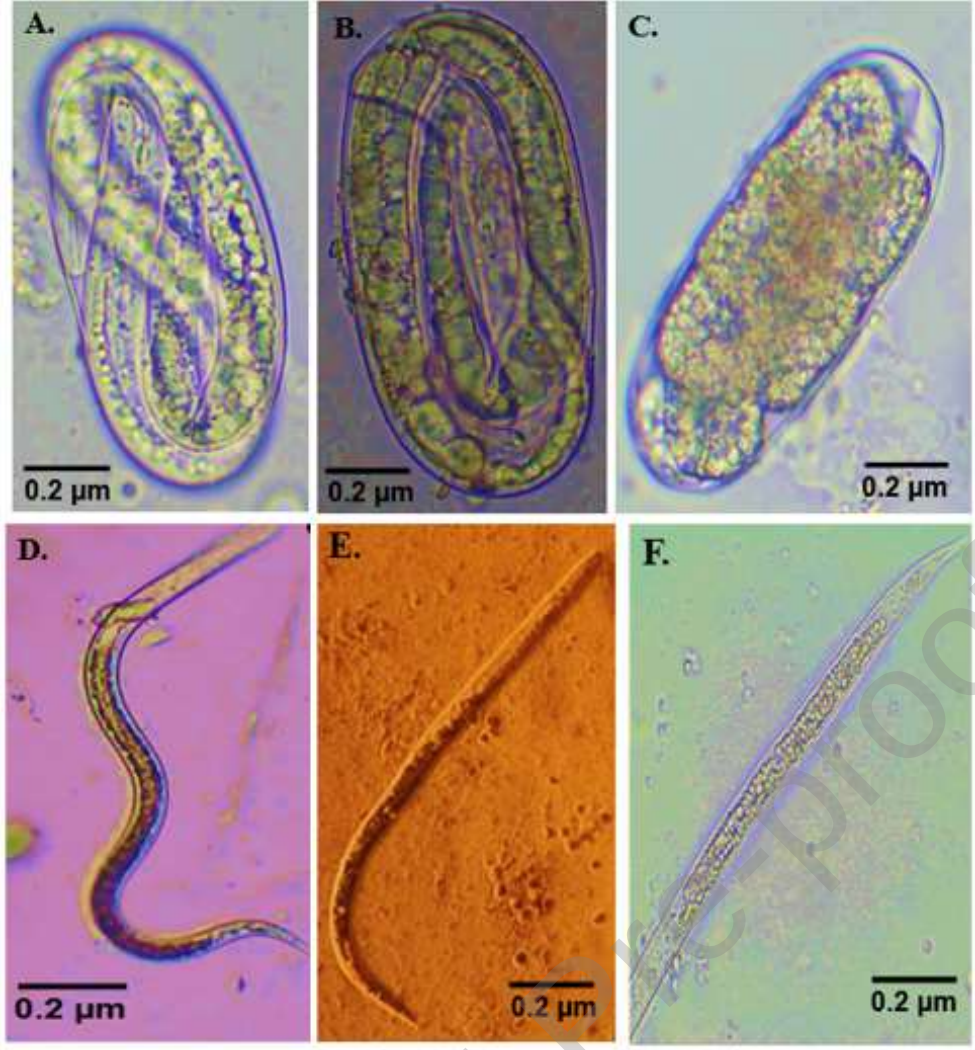

Figure 3. Effect of *Streptomyces* spp. GHRS-5 culture filtrate on *M. enterolobii* eggs and juveniles
(A) Healthy egg of *M. enterolobii* (B) *Streptomyces* sp. GHRS-5 spores infected unhatched egg after
(B) 48 h (C) and 72 h
(D) Healthy juvenile (E) *Streptomyces* sp. GHRS-5 spores infected juvenile (F) Dead juvenile

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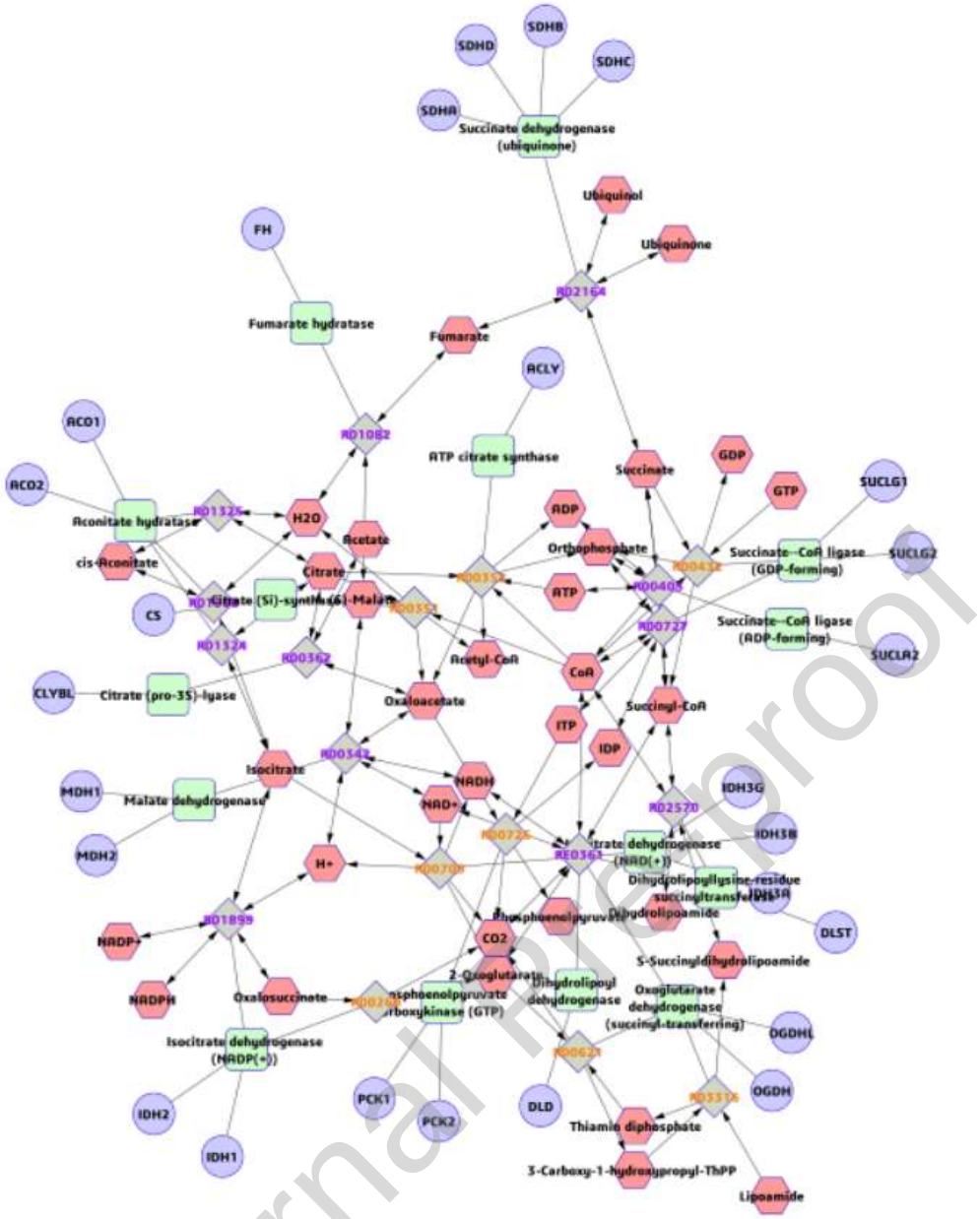

Figure 4.

Pathway enrichment analysis- TCA cycle representing key metabolites, enzymes from effective biomolecules identified from *Streptomyces* sp. GHRS- 5

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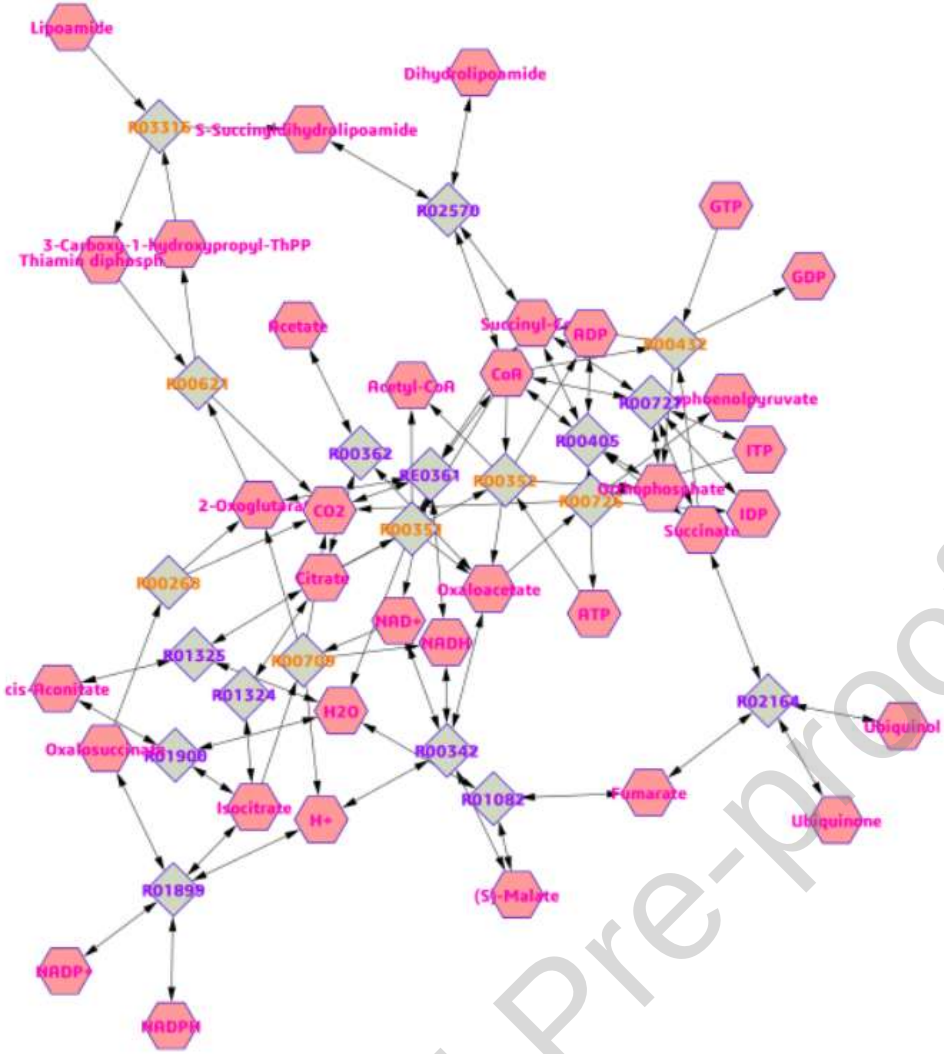

Figure 5. Pathway metabolic network representing the compound reaction metabolism involved

from effective biomolecules of *Streptomyces* sp. GHRS- 5

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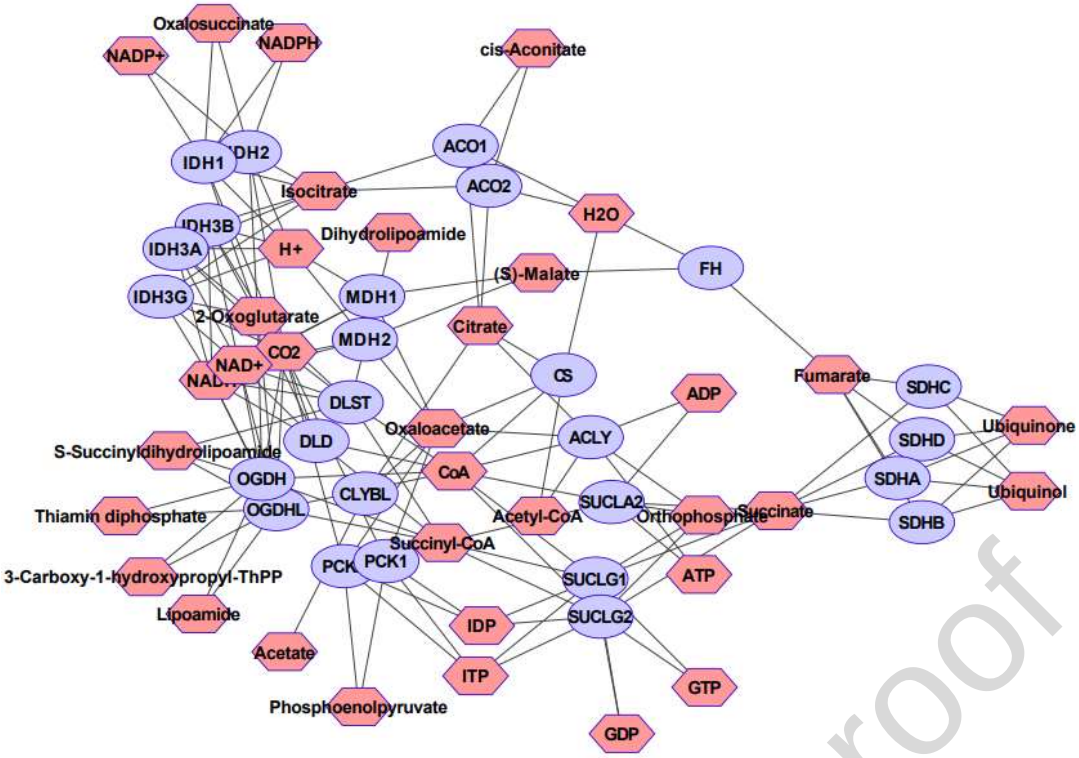

Figure 6. Visualization of the TCA cycle and associated metabolic network by representing the compound-gene-reaction-enzyme present in identified effective biomolecules of *Streptomyces* sp. GHRS- 5

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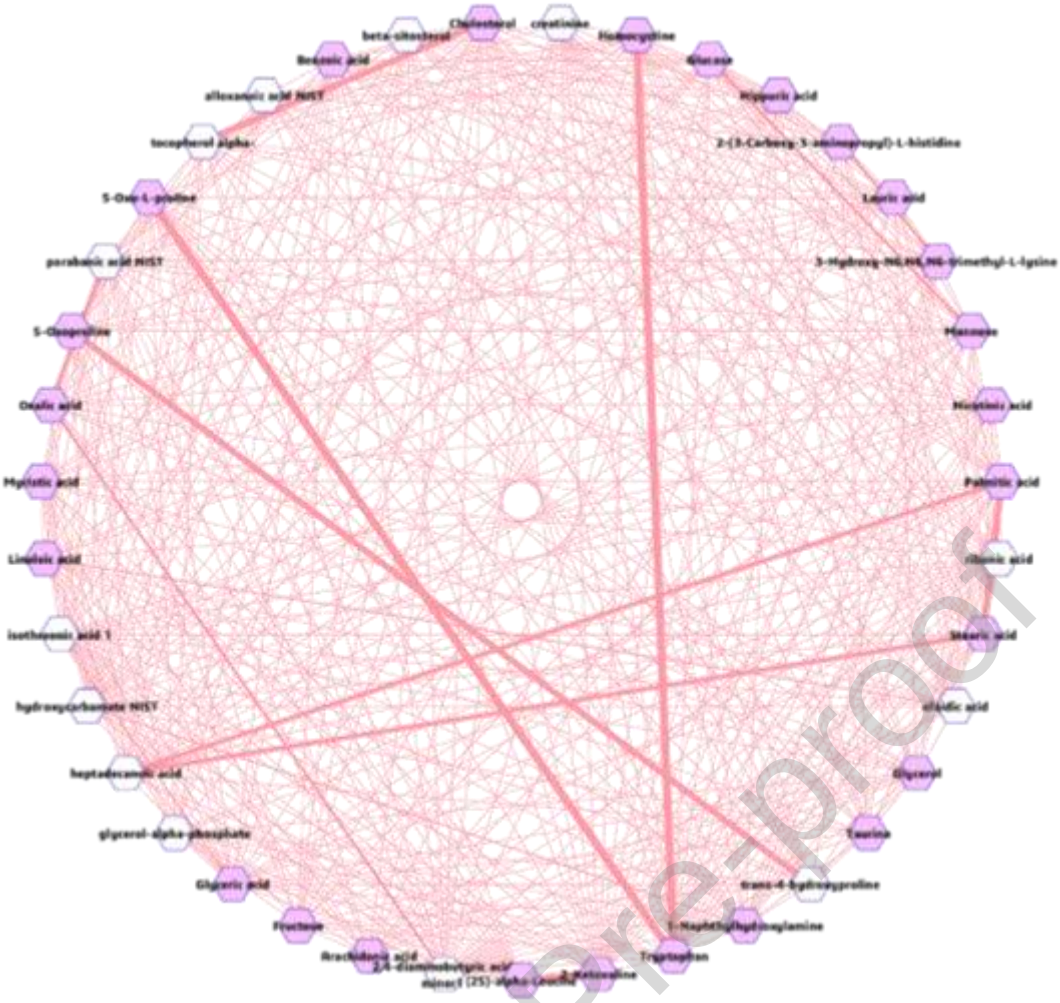

Figure 7. Correlation network diagram depicting interconnections among various biomolecules present in

Streptomyces sp. GHRS- 5 with their standard groups

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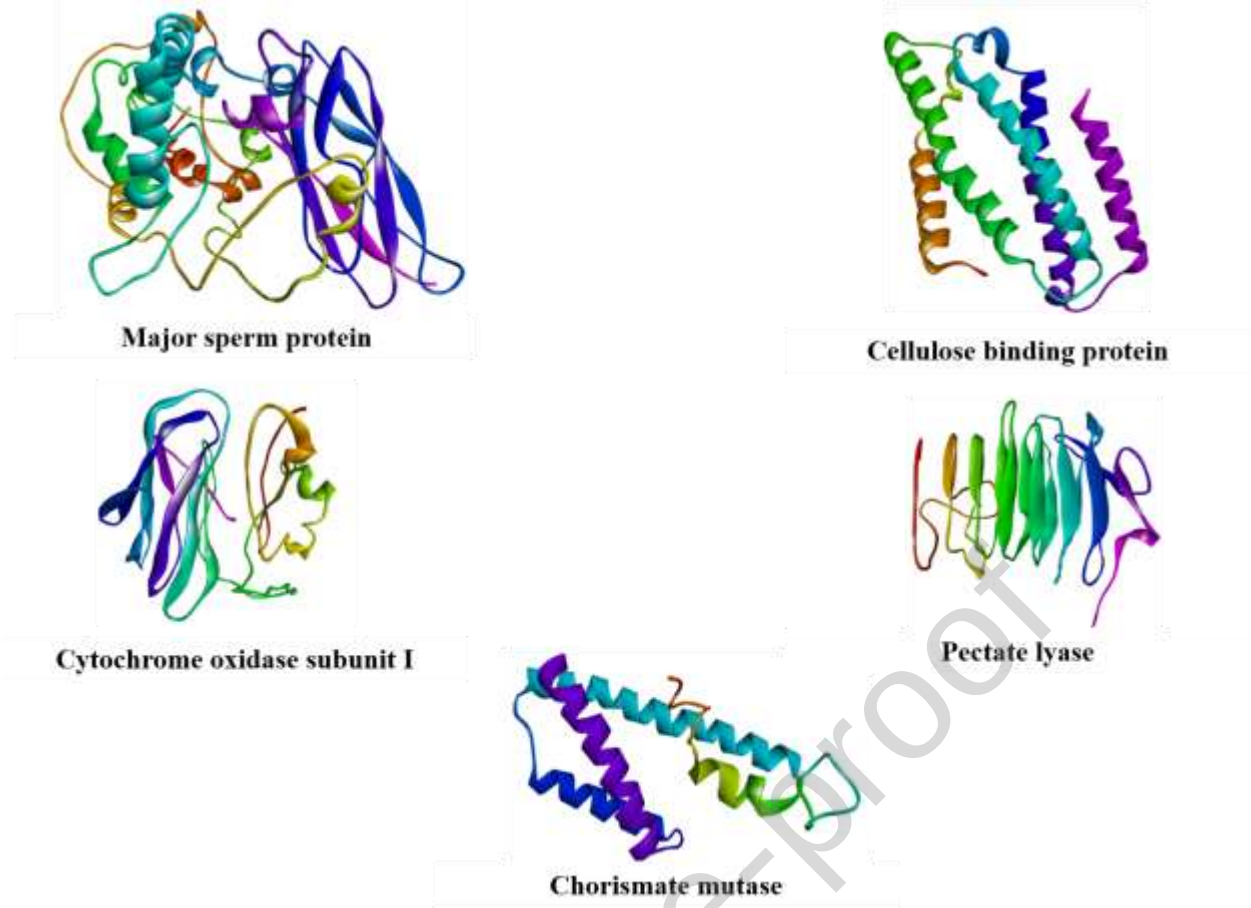

Figure 8. Homology modelled protein targets of *M. enterolobii*

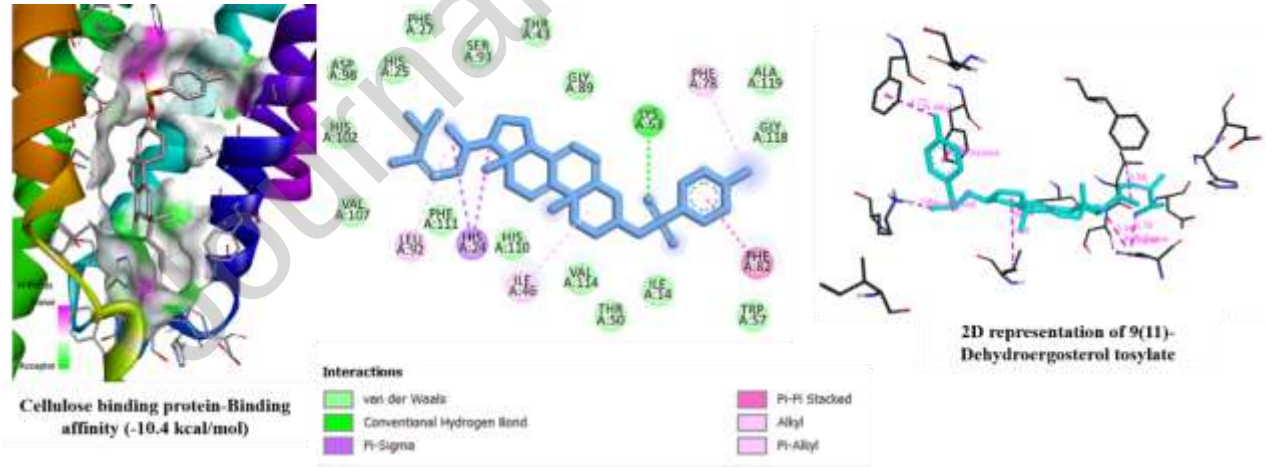

Figure 9. Illustrative diagram representing the protein-ligand interaction of Cellulose binding protein and 9(11)-Dehydroergosterol tosylate

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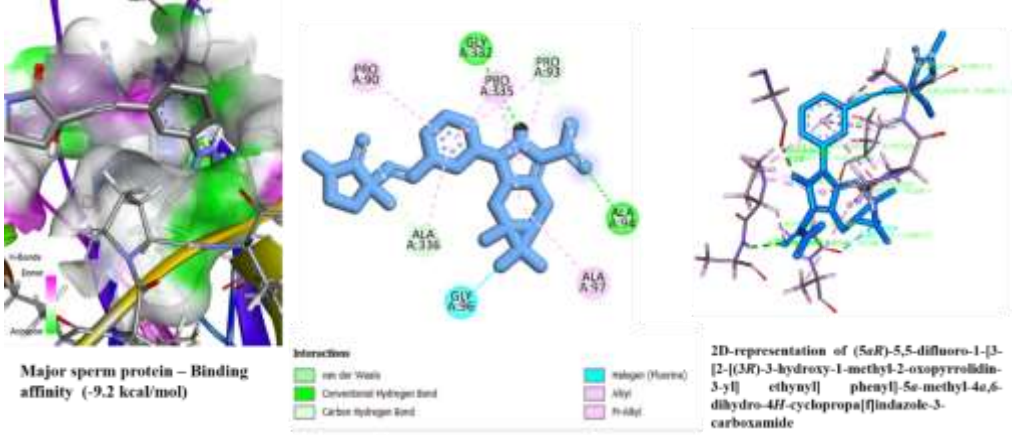

Figure 10. Illustrative diagram representing the protein-ligand interaction of Cytochrome selected protein and (5*aR*)-5,5-difluoro-1-[3-[2-[(3*R*)-3-hydroxy-1-methyl-2-oxopyrrolidin-3-yl] ethynyl] phenyl]-5*a*-methyl-4*a*,6-dihydro-4*H*-cyclopropa[f]indazole-3-carboxamide

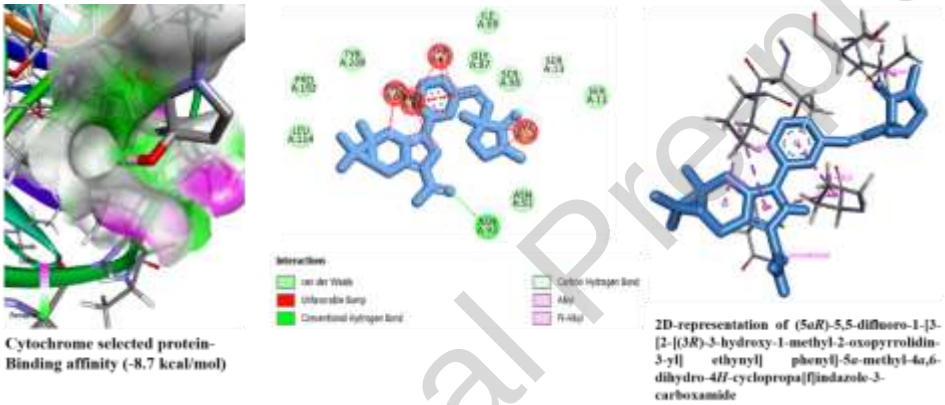

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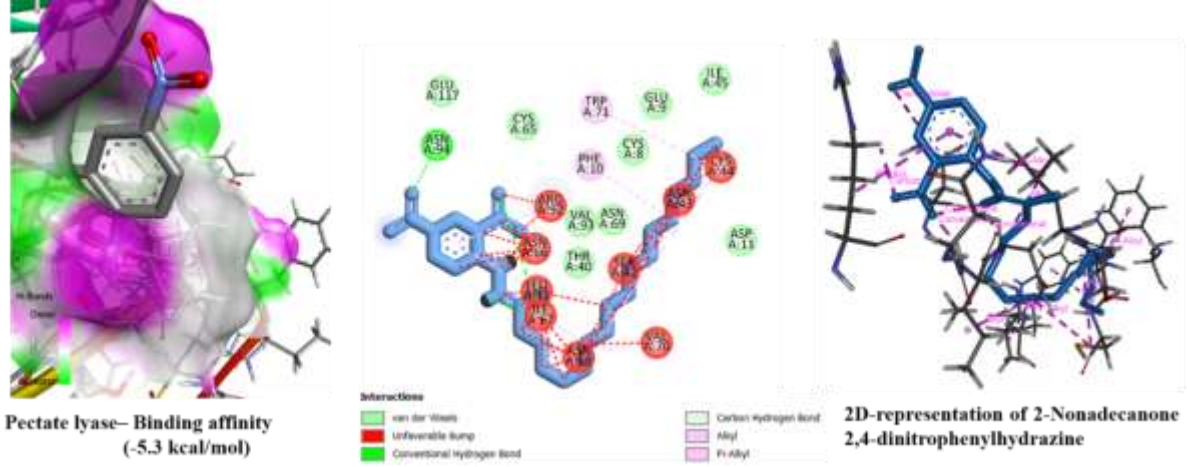
Figure 11. Illustrative diagram represnting the protein-ligand interaction of major sperm protein and (5*aR*)-5,5-difluoro-1-[3-[2-[(3*R*)-3-hydroxy-1-methyl-2-oxopyrrolidin-3-yl] ethynyl]



phenyl]-5*a*-methyl- 4*a*,6-dihydro-4*H*-cyclopropa[f]indazole-3-carboxamide

Figure 12. Illustrative diagram representing the protein-ligand interaction of chorismate mutase and 9(11)-Dehydroergosterol tosylate

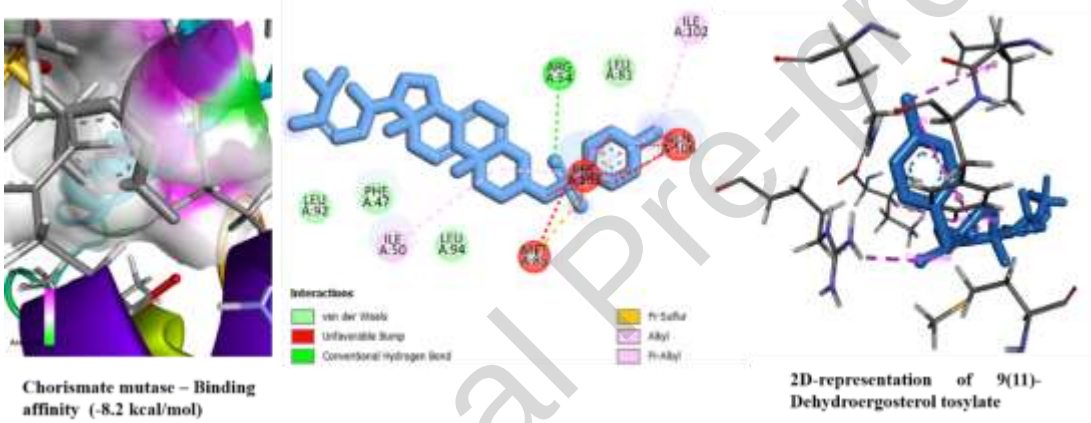


Figure 13. Illustrative diagram representing the protein-ligand interaction of pectate lyase and 2- Nonadecanone 2,4-dinitrophenylhydrazine

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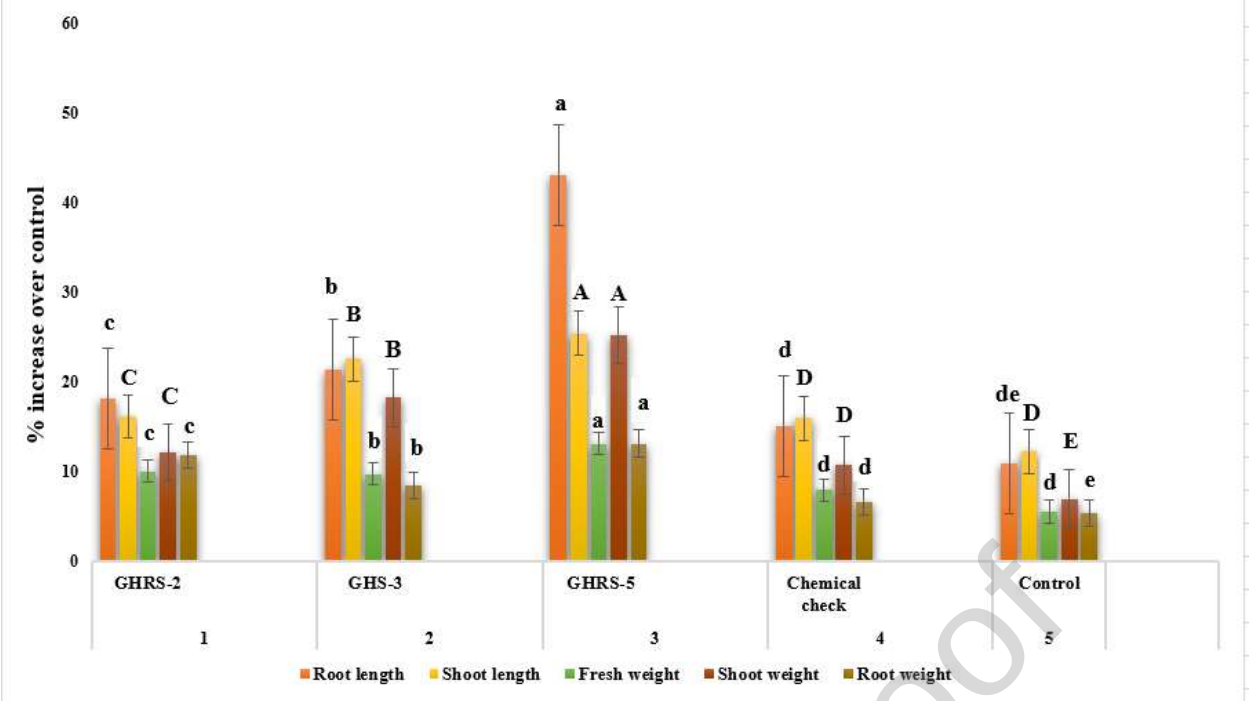

Figure 14. Bio-efficacy of effective *Streptomyces* sp. isolates against *Meloidogyne enterolobii* on guava and their impact on promoting plant biometric parameters. Bars with the same letter indicate no significant as per Duncan's multiple range test ($P < 0.05$).

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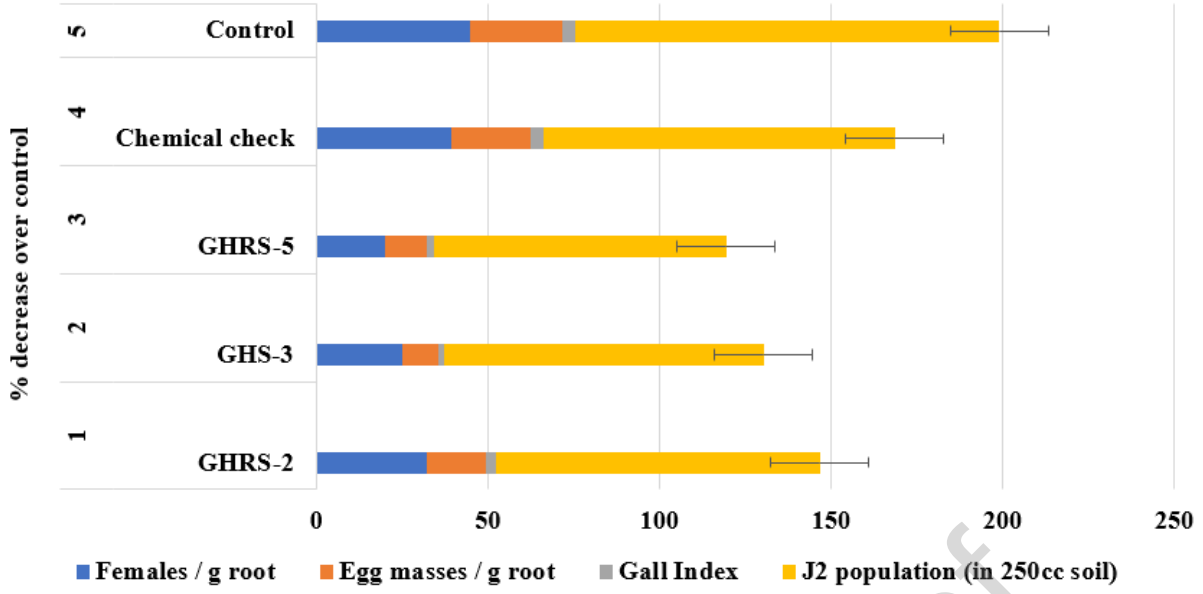

Figure 15. Effect of *Streptomyces* sp. isolates on *Meloidogyne enterolobii* under greenhouse conditions. Bars with the same letter indicate no significant as per Duncan's multiple range test ($P < 0.05$).

Declaration of competing interest

We wish to confirm that there are no conflicts of interest associated with this publication, and no financial support has influenced its outcome. The manuscript has been reviewed and approved by all listed authors, with no additional individuals meeting the authorship criteria. The order of authors has been agreed upon by all contributors. Additionally, we confirm that this work did not involve experimental animals or human subjects. We understand that the Corresponding Author is the sole contact for the Editorial process (including the Editorial Manager and direct communications with the office). My self the corresponding author is responsible for communicating with the other author about progress, submissions of revisions, and final approval of proofs. We confirm that we have provided a current, correct email address which is accessible by the Corresponding Author.

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Highlights

- *Meloiodgyne enterolobii* significantly threatens guava production worldwide
- *Streptomyces* spp. biomolecules effectively suppress nematodes while enhancing plant defenses.
- *Streptomyces* spp. GHRS-5 showed strong inhibition of egg hatching and juvenile mortality under *in vitro*
- GC-MS analysis identified bioactive compounds with nematicidal properties.
- Biomolecules showed strong binding affinities to key target proteins of *M. enterolobii* suggesting suppression mechanisms.
- Combining biocontrol, molecular profiling, and computational studies offers a sustainable solution for nematode management.

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